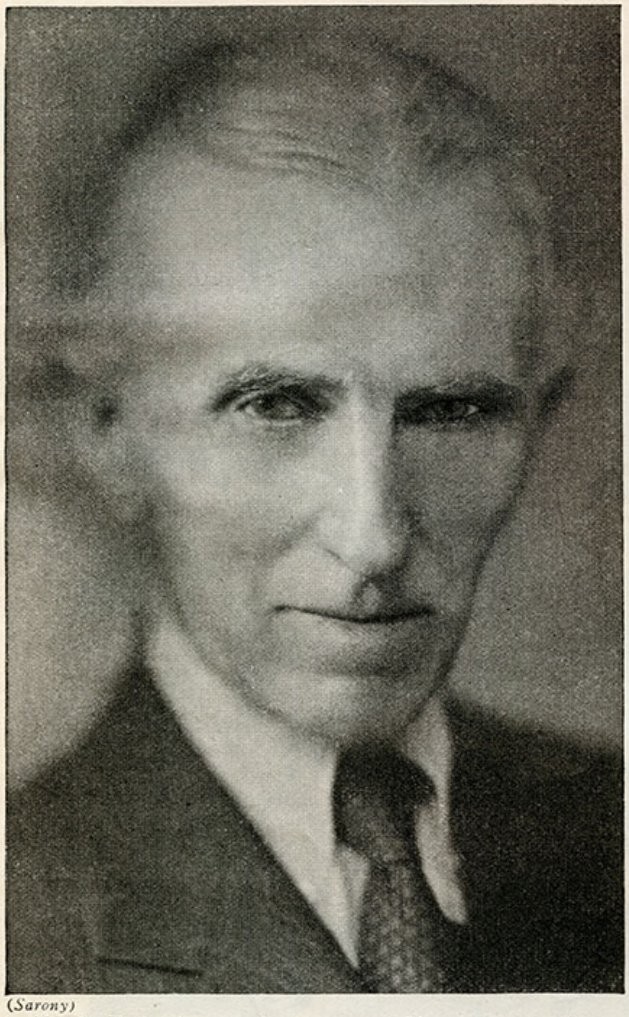
# Nikola Tesla's

*Electrical Wizard, Inventor of Wireless and Power Transmission, Tells His Secrets of Keeping Fit*

*By*

George F. Corners



(Sarony)

Nikola Tesla, keen and vigorous at 78, expects to live more than fifty years longer

LACKING but two years of eighty, I am as fit today as I was at twenty."

Tall, slim, erect, Nikola Tesla was walking briskly across the mezzanine floor of the metropolitan hotel where he lives. There was a sheen in his white hair, and in his blue eyes an unquenchable fire. Time, the great antagonist of all men, has not been able to worst the illustrious inventor in a bout of nearly four-score years.

In that time, Tesla has given to the world five thousand inventions and ideas. "As a matter of history," says C. F. Scott, President of the American Institute of Electrical Engineers, "it is the Tesla principle and the Tesla system which have been the directing factors in modern electrical engineering practise." "His name," adds B. A. Behrend, engineer and author, "marks an epoch in the advance of electrical science." Tesla discovered wireless before Marconi. He solved the secret of high frequency transmissions. His induction motor runs 10,000 ships and plants. Niagara Falls is chained according to his system.

If Tesla's inventions were suddenly withdrawn, half of the transportation systems and half of the industries of the world would stand still. Today at seventy-eight, Tesla still works in three different laboratories. He devotes the major part of his time to the completion of a device which, if successful, will put an end to war. There are innumerable other tricks in the bag of this electric magician.

I was eager to discover how the man who has energized our epoch and helped to make it the Machine Age, recharges the battery of his own body. I met him one afternoon emerging from a long conference, which in turn had followed many hours of laboratory work. But Tesla showed no trace of fatigue. We talked—talked for two hours. At the end he seemed perfectly fresh and wended his way on foot to a special research room of the New York Public Library where he often devotes many hours to his studies.

"How," I asked, "do you find it possible to continue your labors with undiminished vigor at seventy-eight? What is your secret of youth?"

A smile crept over Tesla's features. "It is no secret, but common sense. I look upon the human body as a machine, and I treat it with the respect which a machine deserves. I keep it properly oiled and properly cleaned, and I see to it that it has no opportunity to gather rust.

"The most important element in my scheme of life is my diet. You cannot expect a machine to function well if you do not provide it with the proper fuel.

"Being a scientist, I attempt to regulate the machine which I call my body according to the laws of the planet on which I live. Whatever scheme of life man adopts, it must conform to the physical condition created by the rotation of the globe. The physical law which divides twenty-four hours into day and night also divides man's life into two periods—one in which he lives and another in which he rests. This in itself seems to indicate the desirability of two meals a day in unison with the world rhythm—one meal to give us energy for the day's work, another to supply the body with the material with which it will replenish itself during sleep.

"I have been interested in physical culture ever since I was a young man and have experimented with different nutritional theories. In my early youth I concluded that the ideal way was to absorb food in a fine, almost fluid form and in the proper volume continuously, in small amounts, throughout our hours of work. This method, it seemed to me, would supply steadily energy to the body and impose the least strain upon the digestive organs. Fascinated by the theory, I lived that way for several years, preparing myself dishes con-

# Youth *and* Strength *at* 78

sisting of milk, beef juice, and purées. Working in the laboratory, I took my nourishment whenever I felt the need. My health was excellent, my heart and my digestive organs reacted most favorably.

"However, my work called me away to places where it was impossible for me to prepare my food in the fashion to which I had accustomed myself. I was compelled to take my meals like other men. The change caused considerable digestive disturbances. My intestines and my stomach had shrunk and my interior organs had lost much of their peristaltic power. It took me a long time to adjust myself to the new regimen."

"How many meals a day did you find most conducive to health?"

Tesla laughed: "I have tried everything—six meals a day, like most Europeans; three a day, one a day; I have tried continuous feeding. Finally I decided that two meals are most beneficial.

"Lunch is a superfluous meal, because it is futile in the economy of the human body. Lunch loads the stomach with food which it cannot transform quickly into energy for work. It takes from three to four hours before we can digest our luncheon; three or four hours before the food takes effect. Hence, lunch does not increase our efficiency. On the contrary, the digestive process summons the blood from the brain to the stomach and impedes our mental and physical activities.

"One meal a day does not furnish sufficient energy and can not be harmonized with the movement of the clock which regulates our lives—the earth spinning about itself.

"Two meals can be distributed so as to fit into the natural scheme prescribed by the rotation of the earth. Two meals a day improved my health, sharpened my appetite, and increased the sensitiveness of my palate. I did not know what life was until I eliminated lunch. Today at seventy-eight I enjoy my two meals to the same degree as when I was a boy of sixteen. I do not suffer from dyspepsia or any other ailment. I never find it necessary to take laxatives or other medicaments.

"If my scheme is adopted universally, it will lengthen man's span of life and improve the health of the human race. It is impossible to distribute three meals to advantage. Two meals a day enables the worker to absorb and digest his meal before he begins to work."

"And what is the time which you select for your meals according to your gastro-culinary scheme of existence?"

"The time for eating," Tesla replied, "depends upon climate and conditions of the country. We should take breakfast two hours before we start to work, and we should rest two hours afterward. If a man has his breakfast between seven and eight, he should begin work at ten, work continuously for five or six hours, which would mean that his work day ended at three or four; then he should go home, and devote several hours to recreation, rest and exercise, and take his final meal between seven and eight at night. This arrangement fits in with the cosmic law as well as with the new hours of labor, and it puts the worker in fine condition."

"What is the fuel which you consider most essential for the human machine?"

"Every meal should have some protein and fat, the one for building, the other for burning. Protein builds, fat is a fuel—it energizes.

"In the morning fat should be commended, because fat, taken preferably in the form of butter and oil, is what we need for our work. In the evening I suggest a preponderance of protein to rebuild the cells of the body while it rests.

"I believe with Bernarr Macfadden in the miracle of milk. There are two hundred and sixty grains of protein in one pint of milk. It is easily assimilated. Mr. Ford, unlike Mr. Macfadden, does not believe in milk. He feels it should be artificially manufactured. Unlike Ford, I believe in milk. Ford should take to milk if he wants to turn out more millions of his 'Lizzies.'

"Another excellent ingredient for food is the white of an egg. There are sixty grains of protein in each white of an egg, and only forty grains in the yolk. I prefer to eat only the white. The yolk contains iron and vitamins which are good, but also uric acid, which I avoid as much as possible. Cheese is the most nourishing of all foods, containing one hundred and forty grains of protein per ounce. The next best protein-containing substances are certain cereals which contain thirty to thirty-five grains per ounce of food. Rice carries the least uric acid and can be assimilated most easily.

"A large amount of protein is contained in pulses or leguminous products, beans, peas, lentils, etc. But they also contain no small amount of uric acid. Uric acid is responsible for most ailments which impede the metabolistic processes of the body. It causes rheumatic conditions, arthritis, high blood pressure, and many other maladies of civilized life. Primitive man may have been able to eliminate uric acid thoroughly by hard physical labor. The city dweller should be exceedingly cautious before he introduces large amounts into his system.

"Vegetables are indispensable to any diet, although they contain only six or seven grains of protein per ounce. They do not build and they supply little energy, but they regulate the bowels and counteract uric acid. They also supply the body with the vitamins and essential mineral salts. Moreover, they are convenient carriers for fat. I am fond of a fine, cooked mixture of onions and chopped celery prepared with a great deal of butter. This combination is easily digested. I have in part subsisted on it for many years.

"Fruit, too, is an essential part of any well-regulated diet. It is indispensable, (*Continued on page* 70)

**TESLA is one of our very great men. This is the inspiring story of what his two-meals-per-day plan and a daily ten-mile walk have done for him. Ten miles of walking and two meals a day of the right food will keep any man still young and strong at Tesla's time of life.**

**Some of Mr. Tesla's views do not conform to our own in every detail, but are the result of his personal experiences, and our readers are entitled to have the benefit of his views first hand.**

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## ¶ *Nikola Tesla's Youth and Strength at 78* —

Continued from page 23

partly because of its bulk which aids the digestion, and partly because of its alkaline reaction. It is a well-known fact that even citrous fruits, which have an acid reaction at first, change to alkaline in the processes of digestion. The alkaline are the natural enemies of the acids. Sugar in small amounts is not objectionable. There is, however, plenty of sugar in fruit."

"Are you a vegetarian or a carnivorous animal?"

Tesla replied, "I eat meat only at rare intervals, possibly only once or twice a year. It is rich in proteins but also in uric acid. I prefer fowl to beef. Fish is more digestible and almost equally rich in proteins and in fats. The phosphorous it contains is excellent for the nervous system. But phosphorus, while feeding the brain, causes acidity. Both meat and fish give us quick energy. Under certain conditions, for certain workers, they are indispensable. They shorten life by introducing more uric acid into the system than it can conveniently eliminate. But while we live they give us considerable vigor."

"AND what," I asked, "is your attitude toward alcohol?"

"Since prohibition was enacted I have not touched a drop, and long before that I gave up all other stimulants. I gave up alcohol, in spite of the fact that it was beneficial to me, in obedience to the law, and I find that I cannot now return to it. I had previously expected to live 150 years. Now that I have given up alcohol, I have reduced my expectancy of life to 135 years. In Yugoslavia, where I was born, alcoholic drinks are taken regularly with meals. That habit rarely leads to drunkenness.

"Alcohol has helped me a great deal in my life. It has made possible for me to escape from a strain which otherwise might have proved fatal. The man who is not an inventor cannot realize how powerfully the entire system may be affected by the discovery of a new truth. When I first saw my own invention, the induction motor, at work, my heart nearly gave out. Fortunately a bottle of brandy was near at hand of which I partook. This balanced my nervous excitement, and I was able to survey the rotation of the motor with philosophic poise.

"I was strongly addicted to coffee, with the result that my heart was subjected to a dangerous over-strain. It took me eleven years to vanquish my craving for this pernicious drink. The cure cost me a good deal of money. For eleven years I had the finest coffee prepared for me at my table every day, and never touched a drop. Its presence aroused my powers of resistance. If it had not been there I would have suffered from the craving. I, at least, have found coffee and tea rank poisons."

"You have not," I interrupted. "expressed your opinion on beer."

"I have cared less for beer than for wine or brandy. As a young student I drank beer moderately, about sixteen seidels a day, whereas most of my fellow students consumed sixty. Beer gave me a katzenjammer. It is drunk

in large quantities because for some reason it is easily absorbed. No one can drink twenty or thirty glasses of water or milk, but it is not difficult to drink thirty or forty glasses of beer. Owing to the large volume he consumes, the beer drinker absorbs more alcohol than the whiskey drinker. Moreover, the tremendous amount of liquid which he gulps down enlarges his stomach and increases his girth. Some people seem to thrive on beer, but they usually acquire enormous bellies.

"And what," I asked, "is your attitude toward smoking?'

Mr. Tesla's defense of alcohol as against coffee and tea had left me somewhat startled. However, it seems that Tesla is no friend of the weed.

"I have smoked a good deal, but when I was about twenty I gave a promise to stop, and smoked my last cigar. Smoking stimulates thought on general subjects; but is an enemy to concentration. The most stimulating activity, to my mind, is walking. I walk never less than ten miles a day. This has kept me in good health, but it may in the end lead to my destruction, because I am a confirmed jay-walker.

"My exercise," Tesla continued, "is not confined to my ten-mile walk. I bathe every day and I exercise in my bath. I have acquired great proficiency in these exercises. I have, in some respects, the agility of a baby. There is hardly a movement I cannot make except I cannot kiss my elbow. I can clamp a piece of paper between my shoulder blades and hold it so firmly that no one can remove it without tearing it off."

"It seems curious," I said, "that you, who are one of the master minds in electricity, do not use it to recharge your own energies."

"Why, of course I do," Tesla replied. "Many years ago I discovered the therapeutic uses of electricity, especially of the high-frequency current. I announced my discovery in this field in 1891.

"I believe in what may be called a waterless bath, by which I mean charging of the body to a very high electric potential. It is a bath of fire that rebuilds, rejuvenates, cleans, and exhilarates. It carries off instantly all dust, impurities and microbes, and stimulates the tips of the nerves. While in the bath the body is surrounded by a halo of light, plainly visible in the dark. I am now working on an apparatus which will make this electric bath safe and economically possible, even for a person of average means."

TESLA, unlike Edison and Marconi, is a bachelor. He prefers to be wedded to science. It seemed all the more important to discover his attitude on sex—sex, the most electrifying factor in the lives of most men. Tesla did not evade the question.

"I recognize the importance that sex plays in the life of man. Nature has made its attraction irresistible to insure the perpetuity of the race. As for myself, I have found that the thinker is confronted with the problem of perpetuating either the species or the mind. It is almost impossible, at least in certain realms of high endeavor, to do both. This has been recognized by the Hindus, whose adepts practise complete sexual abstinence, and by many of the great religions.

"Before I produced the rotating magnetic field, I concentrated all my powers upon my experiment. The strain would have killed a hundred oxen. I

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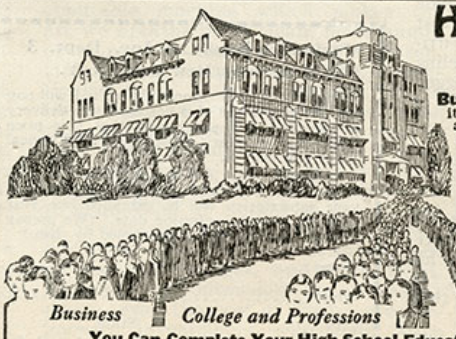


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## Are Burdened With the Scourge of

# Constipation

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And so the vicious circle continues—the miserable sufferer changing from one drug to another, seeking for something that will bring about the desired cure—and with each new drug making his condition worse.

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Ever since the days of Hippocrates, writers on medical and hygienic subjects have emphasized the harm that results from constipation.

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The man who is constipated is a poisoned man. His intestinal tract is a factory of infection, making and liberating millions of active and dangerous disease germs.

If allowed to become chronic, constipation may result in any of the following diseases. Bright's disease, catarrh, consumption, diabetes, dysentery, kidney trouble, liver complaints, neurasthenia, pneumonia, rheumatism, stomach troubles, coughs and colds, vital depletion, neuralgia, insomnia, dyspepsia, and heart irregularities.

Constipation brings on a derangement of the nervous system. It brings on piles and other troubles that cause the most intense suffering.

The chronic constipate is usually pale and nervous. He lacks ambition and zest for work. He is often depressed and irritable. And there is no reason for his feeling otherwise. A constipated body is a clogged body—alive with poisons that drain it of health, vitality and ambition.

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certainly could not have survived it if I had permitted my energies to be diverted into the channels of sex."

"You have not," I remarked, "told me the part played in your life by that great restorer of energy—sleep." I asked this with malice aforethought, because I remembered that Tesla's great rival, Edison, had always claimed that he was able to get along with only a few hours of sleep.

"I should say," Tesla replied, "that a man who performs hard physical labor needs seven to eight hours of rest to recuperate, not necessarily eight hours of sleep. I myself sleep only two hours out of twenty-four. This does not mean that I may not assume a recumbent position, but my mind continues to work out its problems. I have often, during such apparent periods of rest, worked out my most difficult problems. In my younger days, I probably slept three to four hours.

"When I sleep, I sleep profoundly. To sleep deeply, profoundly, is an art like deep breathing that must be learned. It is one of the secrets of the East. I think that I have discovered the secret. After a profound sleep, no matter how short, I feel so much refreshed that even now, when I am near eighty, I must perform gymnastic feats to subdue my new-gained vitality."

"EDISON," I here interjected, "declared that he could get along on two or three hours of sleep."

Tesla's eyes twinkled merrily.

"People sometimes deceive themselves," he said. "Edison thought that he slept very little. As a matter of fact, I never knew anyone who slept as much as Edison. I have often seen him put his feet on the table in his laboratory and sleep for three or four hours. Then he would get up and work a little, only to take another nap of three or four hours. Naturally he did not have much need for sleep when the day was over."

"You would not," I remarked, "recommend two hours of sleep or the extreme indulgence in brandy to other men."

"It is possible," Tesla replied, shrugging his shoulders, "that my scheme of life may not help others unless it is modified in accordance with one's specific requirements, but it has kept in gear for seventy-eight years the machine which is Nikola Tesla."

## ¶ *Now We'll All Be Air-Conditioned*

Continued from page 15

modern system. The whole point in modernization is to make use of the possible but to do away with the costly and inefficient makeshift.

Several companies offer a portable air conditioner which may be put into operation in one room at a time. The cost of these units for a number of rooms is prohibitive for all but the very wealthy, but their benefits to the average customer are manifold. They are sightly, no more obtrusive than the ordinary radiator, and can be bought in any finish matching usual present-day woodwork. And they can be